

SHITI GHOSH

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EDUCATION

Master of Science (M.S) in Bioinformatics (GPA:4.0/4.0)

Sep 24 – May 26 (Expected)

College of Science, Northeastern University, Boston, USA

Prominent Coursework: Statistics for Bioinformatics, Bioinformatics Programming using Python, Computational Methods in Bioinformatics, Genomics

Master of Technology (M.Tech.) in Biotechnology (GPA:3.7/4.0)

Jul 15 – Jun 17

School of Biotechnology and Bioinformatics, D. Y Patil (Deemed to be) University, Mumbai, India

Bachelor of Engineering (B.E) in Biotechnology

Jun 10 – Jul 14

Manipal Institute of Technology, Manipal University, Manipal, Karnataka, India

TECHNICAL SKILLS

Programming Languages	: Python, R, Unix/Linux Shell Scripting, SQL, PL/SQL
Bioinformatics	: FastQC, Trimmomatic, Trim Galore, MultiQC, DESeq2, kallisto, HTSeq, featureCounts, Salmon, BWA, STAR, HISAT2, GATK, Samtools, SEURAT, Scanpy, GEO, TCGA, SRA
Computational Tools	: High-Performance Computing, Git Version Control, Jupyter Notebook, Docker, Singularity
Biotechnology	: Mammalian cell culture, Atomic Force Microscopy, PCR
Project Management	: Agile, Scrum, Waterfall
Design	: Adobe XD, Figma
BI/Analytic Tools	: R Shiny, Tableau, PowerBI, R markdown

ACADEMIC PROJECTS

Bulk RNA-Seq Analysis Pipeline Development

Dec 24 – Jan 25

- Developed and implemented a streamlined RNA-Seq analysis pipeline to process and analyze Next Generation Sequencing (NGS) cancer datasets, optimizing workflow management for consistent and reproducible gene expression analysis.
- Automated the process with Makefile for HPC environments, ensuring scalability, quality control, alignment, quantification, and clear result visualization through analytical plots.

Interactive RNA-Seq Analysis and Visualization with R Shiny

Nov 24 – Jan 25

- Developed a comprehensive RNA-Seq analysis pipeline for mouse using Kallisto for transcript quantification and executed it on high-performance computing systems, ensuring scalability and optimized performance.
- Created an interactive R Shiny dashboard embedded within an R Markdown document, allowing dynamic visualizations (PCA plots, volcano plots, heatmaps) and KEGG-based enrichment analysis, while integrating biomaRt for querying Ensembl for gene annotations to enhance the biological interpretation of RNA-Seq data.

Analysis of Single-Cell RNA Sequencing Data

Nov 24 – Jan25

- Performed quality control, normalization, and identification of highly variable genes using Seurat, ensuring biologically significant features for analysis.
- Conducted PCA, UMAP, and clustering to categorize PBMCs into distinct gene expression profiles, providing insights into cellular heterogeneity.

PROFESSIONAL EXPERIENCE

Technology Consultant, Big Faction Consultancy Pvt. Ltd., Varanasi, Uttar Pradesh, India

Jan 22 – May 22

- Provided observations and actionable recommendations on potential risks in client projects by thoroughly understanding their processes and conducting stakeholder interviews, leading to improved project outcomes.
- Analyzed data from Oracle databases, cleaning and querying information to develop visual dashboards that displayed key performance indicators (KPIs), empowering leadership teams to make informed decisions, resulting in a 5-10% improvement in business performance.
- Collaborated with leadership and cross-functional teams to gather business requirements, design user-friendly websites using Adobe XD and Figma, and align design efforts with business objectives through research, usability testing, and gap assessments, ensuring a seamless and intuitive user experience across platforms.

Research Associate, Happie Hemp Pvt. Ltd, Bengaluru, Karnataka, India

Mar 21 – Dec 21

- Collaborated with Ayurvedic doctors and research teams to integrate traditional herbal knowledge with data-driven insights, supporting the development of hemp-based wellness products while optimizing market strategies through Python-based data analysis, driving a 15% increase in customer engagement.
- Managed project milestones, requirements, and risk management, ensuring the on-time delivery of initiatives, while also spearheading the creation of impactful creative decks that increased brand visibility and audience interaction.
- Led the development and launch of a telemedicine platform in partnership with Ayurvedic doctors, enabling remote consultations and prescriptions for hemp products, and ensuring full regulatory compliance for the platform's operation.